ContinuityGuard

Theme 1: Real-Time AI for Critical Decisions / System Anomaly Detection

Warn before disruption. Inspect the evidence. Preview the next step.

Native LinuxONE AI + accountable operator decisions

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Public code: github.com/ANBU-SELVAN-1907/RASCALS-IBM-DATATHON

Problem statement

Identify abnormal system activity as it happens and alert organizations before a major disruption. Target users: hospital IT operators.

1,294+

outpatient appointments postponed

NHS England reported this disruption across two affected trusts for 10-16 June 2024 after the Synnovis cyber incident.

Source: NHS England, 20 June 2024

Context only: our prototype was not tested on that incident and does not claim to detect or prevent ransomware.

The operational problem

Pool exhaustion, retry cascades and resource pressure can spread through shared dependencies before operators know where to investigate.

The decision we support

Which dependency needs attention, which software services matter most, and what should an operator inspect next?

Solution: a warning-to-decision pipeline

The browser uses TLS or SSH. Telemetry, inference, local generation and the journal stay in the LinuxONE VM.

JUPYTER CONTAINER / NATIVE s390x

Fit candidates > separate calibration > validation policy > fresh final test > hashed model

Ingest

Validated queue

Observe

Causal features

Score

Forecast + guard

Decide

Evidence + runbook

ISOLATION FOREST

Separate novelty score; not a probability or independent page trigger.

LOCAL GRANITE

Verified evidence wording; complete IDs and labeled fallback.

RESPONSE PREVIEW

Same-workload action/no-action branches in the simulator only.

A small model, selected for the actual task

Predict a new sustained service-objective breach within 60 event-time seconds using current and past observations.

60
Train runs

18
Calibration

24
Validation

60
Fresh test runs

Calibrated boosted trees

Seven-leaf histogram boosting won the validation objective. Current policy: forecast >= 0.97 OR explicit operational thresholds.

Evidence of separation

26 causal features; whole runs and disjoint seeds. No scenario, onset, future values or fault labels enter the predictors.

Forecast PR-AUC 0.786 / prevalence 3.8% / Brier 0.01601

v3 aligns incident matching to 60 s. Earlier v1/v2 reports are retained as development history.

A working operator console, not a mockup

Actual deployed s390x application. Scenario telemetry and response effects are synthetic.



Retained warning evidence

Connection pool occupancy

83.3 %

Waiting queue

12.3 requests

Memory occupancy

37.3 %

Actual selected incident; measurements are evidence, not causal attribution.

Streaming services > persistent warning > measurements > declared dependency and runbook

Measure usefulness without hiding the misses

Fresh v3 test: 60 synthetic runs; correlated service incidents within a run. No production hospital validation.

Detector	Early recall	Precision	Median lead*	Late	Missed
Static thresholds	72.1%	60.0%	10.0 s	2	10
Robust statistics	44.2%	11.9%	39.0 s	0	24
Isolation Forest	58.1%	17.7%	41.0 s	0	18
ContinuityGuard	**72.1%**	**59.3%**	**13.0 s**	**1**	**11**

Both methods warned before 31/43 incidents. Median lead: 13 s vs 10 s; model precision was slightly lower. This does not establish consistent ML superiority.

Run-bootstrap 95% interval for early recall: 58.0%-88.6%. 500 resamples; simulator-conditional.

*Among early detections only. Zero healthy-run alerts in 10 service-hours is not proof of zero real false alarms.

The response is inspectable and bounded

Evidence, service priorities, approved investigation steps and a reproducible response preview form one decision workflow.

ACTUAL SAVED APPLICATION OUTPUT

database needs operator investigation. Investigate database connection pressure. Supporting measurements: Connection pool occupancy 83.3 % (E1); Waiting queue 12.3 requests (E3); Memory occupancy 37.3 % (E4).

+315 successful urgent-results transactions

Modeled breach: 104 s without action to 0 s with action. Tradeoff: 1238 optional requests deferred.

Source: evidence/demo-verification.json / synthetic scenario

Predictive + generative, locally verified

Granite generated schema-validated wording on s390x in 15.81 seconds in the Jupyter proof. Every evidence ID cited in the summary is retained.

The preview compares allowlisted actions under identical future demand/noise. It reports benefits with deferred optional work. It executes no real remediation.

Suspected source is a hypothesis, not causal proof.

Platform use that judges can verify

Native AI execution, actual ingestion and container controls are exposed as evidence.

37

native tests passed

Checks cover leakage, online/offline parity, time gaps, invalid inputs, provenance, overload, consumer failure, model corruption and grounded fallback.

1000/1000 events processed

Real local HTTP path: 100 offered events/s for 10 s. Queue-ack p95 8.006 ms; queue wait and inference are distinct boundaries.

Application containment

UID 1000 / read-only root / capabilities dropped / no-new-privileges / 2 GiB memory / 1.5 CPU quota / Authenticated TLS / SSH


```
{
  "project_title": "RASCALS_IBM Z Datathon 2026 Project",
  "architecture": "s390x",
  "python": "3.12.15",
  "process_uid": 1000,
  "cgroup_memory_max": "2147483648",
  "cgroup_cpu_max": "150000 100000",
  "effective_capabilities_hex": "0000000000000000",
  "no_new_privileges": "1",
  "root_filesystem_readonly": true,
  "model": {
```


Actual live proof excerpt. Full screenshot and model hash are in evidence/. No accelerator claim.

Scenarios: normal change is not always failure

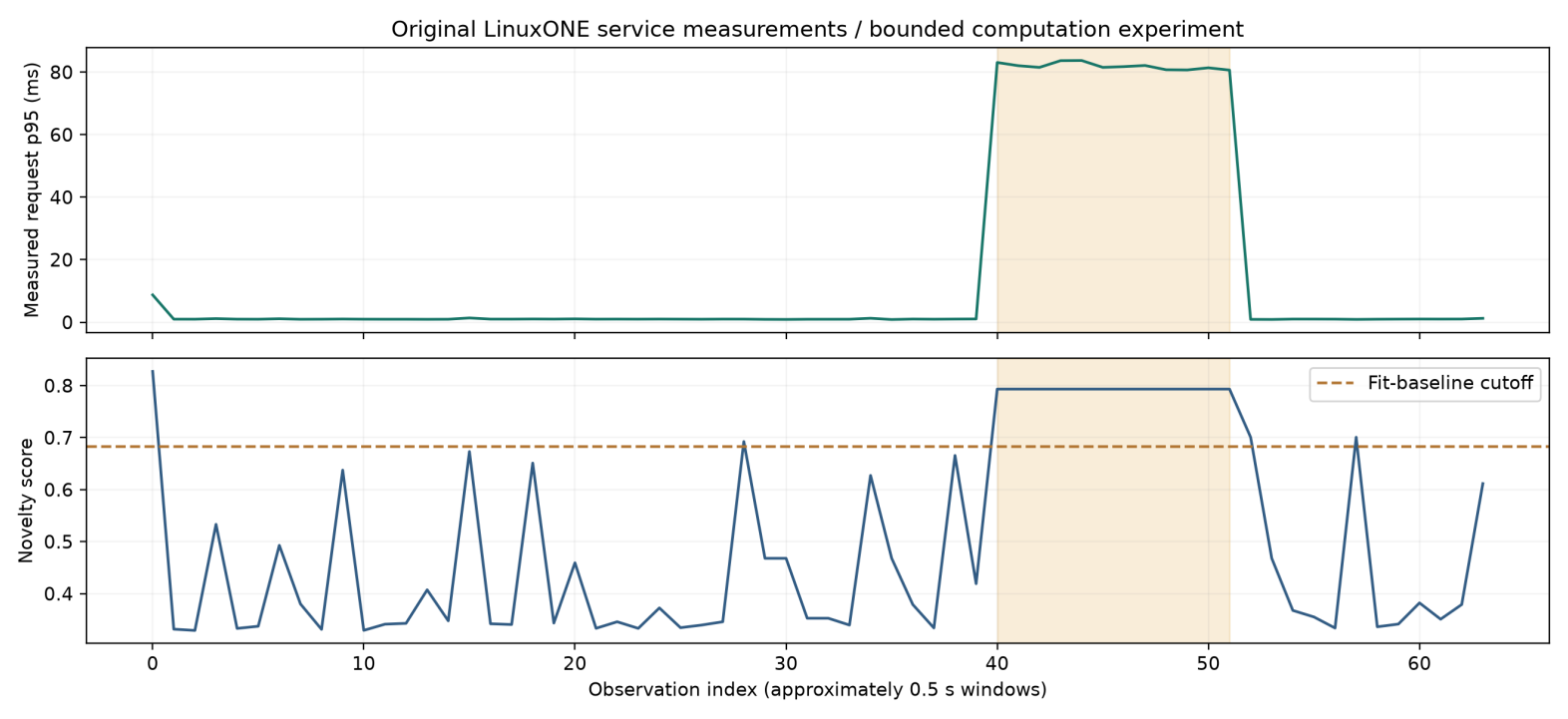
Fixed seed 101 / severity 1.12 / frozen v3 model. These are synthetic illustrations, not a new independent test.

Scenario	All-service alerts	Urgent warning	Urgent breach
healthy	0	None	None
healthy surge	0	None	None
healthy shift	0	None	None
pool exhaustion	4	167 s	188 s
retry storm	5	182 s	219 s
memory pressure	1	None	None
abrupt failure	4	94 s	94 s

Pool and retry examples warn before sustained breach. Abrupt failure is detected at onset. The memory example alerts another service; it does not invent an urgent-service breach.

Original data: measured on this LinuxONE VM

64 actual service observations / bounded request computation / raw CSV, timestamps and SHA-256 retained. No patient or hospital data.



Observed baseline check

0.98 ms

Controlled compute pressure

81.64 ms

Median measured request p95 across windows. All 12 pressure windows exceeded the fit-baseline novelty cutoff; 2 recovery windows also did.

Separate small novelty experiment. It does not validate the simulator-trained breach model, production recall or real outage prevention.

What makes the integration distinctive

The useful output is an inspectable operational decision, not an unexplained anomaly score.

01

Retain the warning evidence

Measurements and declared downstream priorities remain attached to each incident.

02

Constrain local explanation

Granite selects verified wording and complete evidence IDs; invalid generation is explicit.

03

Preview the tradeoff

Compare allowlisted action and no action under the same synthetic future workload.

Established algorithms; original project integration. No world-first or autonomous-prevention claim.

Learnings during the datathon

Changes we can explain from actual implementation, tests and corrected evidence.

01 Platform reality

RHEL, Podman and native s390x packages required different setup from Ubuntu examples.

02 Evaluation alignment

A 60-second prediction needs a matching horizon; correcting it reduced the headline.

03 Timing boundaries

Queue acceptance, inference and completed processing are different measurements.

04 Grounded generation

Every evidence reference must match the generated wording; fallback must be labeled.

05 Data provenance

Measured request behavior, controlled simulation and production hospital data are different.

Implementation and IBM Toolkit fit

Five native Jupyter notebooks, a deployed service, and a documented optional serving path.

01 / Prepare

Provision the VM, complete assigned labs and retain persistent Jupyter storage.

02 / Train and verify

Fit/calibrate/select, then review fresh final results and engineering tests.

03 / Measure and deploy

Collect original service measurements; build the resource-limited app and verify its journal.

04 / Demonstrate

Open authenticated TLS by IP, inspect an alert and explain/preview the next step.

IBM AI Toolkit

Applied deployment guidance: separate functions, non-root app, read-only root, no-new-privileges, TLS and authentication.

Snap ML preparation: a separate supported-family Random Forest candidate was trained and ONNX-checked on s390x. It is not the deployed model.

Registry access is unavailable. Supported Toolkit container execution and AI acceleration are not claimed.

An accountable path from warning to action

Original integration and measured implementation, with limits that operators and judges can inspect.

Community value

Earlier investigation of shared service failures; preserve attention for essential software services.

Engineering evidence

Native notebooks, actual API ingestion, model hashes, failure tests and local explanations.

Next validation

Authorized real telemetry, operator feedback, domain calibration and reviewed runbooks before operational use.

github.com/ANBU-SELVAN-1907/RASCALS-IBM-DATATHON

Both sample labs executed before project work; confirm organizer mapping to the required categories. Review/upload the <=3 minute video and submit the portal manually.

AI assistance is disclosed. Open-source resources and a criterion-by-criterion evidence map are in the repository. No score or award is guaranteed.